torch.signal.windows.hann#

https://pytorch.org/docs/stable/generated/torch.signal.windows.hann.html

Description:

Computes the Hann window. The Hann window is defined as follows: The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window. sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

Function Signature

```
torch.signal.windows.hann(M, *, sym=True, dtype=None,
layout=torch.strided, device=None, requires_grad=False)
[source]#

Parameters

Keyword Arguments

Return type
```

Parameters

Keyword

sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Defaul

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric Hann window.
>>> torch.signal.windows.hann(10)
tensor([0.0000, 0.1170, 0.4132, 0.7500, 0.9698, 0.9698, 0.7500,
0.4132, 0.1170, 0.0000])

>>> # Generates a periodic Hann window.
>>> torch.signal.windows.hann(10, sym=False)
tensor([0.0000, 0.0955, 0.3455, 0.6545, 0.9045, 1.0000, 0.9045,
0.6545, 0.3455, 0.0955])
```

Detailed Documentation

Documentation

Computes the Hann window.

The Hann window is defined as follows:

The window is normalized to 1 (maximum value is 1). However, the 1 doesn’t appear if M is even and sym is True.

M (int) – the length of the window. In other words, the number of points of the returned window.

sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.